the mass of an electron and the mass of a proton, and which are associated somehow with these nuclear forces. Any further clarification will become possible only when the very high energy nuclear disintegration processes have been investigated much more thoroughly than they have been up to now.

III. THE SATURATION OF NUCLEAR FORCES

The above-mentioned analogy with valency forces suggests a conclusion which, even though we cannot prove it here, we can at least make plausible. An essential difference—among others—between valency and electric forces is that the former are capable of saturation. The chemist provides the symbol of every atom with a certain number of valency bonds, the number of which corresponds to the valency of the element. In every structural formula of a saturated chemical compound, such valency bonds are represented by lines issuing forth from the atomic symbol and terminating in another atomic symbol, from which, in turn, as many such lines issue forth as the numerical value of its valency number. For instance, carbon dioxide, O=C=O, the compound of the tetravalent carbon atom and two divalent oxygen atoms. The characteristic feature in this case is that an atom, all the valency bond lines of which are taken up, is saturated, in other words, it has used up its valency. Thus, for instance, in the case of the water molecule, H—O—H, the two valencies of the oxygen atom are saturated by hydrogen atoms, and no further hydrogen atoms can be bound. A more or less normal molecule of the OH_3 type is non-existent.

Now the nuclear forces or at least a large part of them have a completely similar property of saturation. A neutron can bind to itself not more than two protons, and a proton not more than two neutrons. If we wished to express this fact in the form of valency lines, the proton would have to be written with two such lines, which could connect with neutrons only, and similarly, a neutron would have two valency lines, which could connect with protons only. This is, of course, not quite correct, since we are disregarding the forces which act between any two protons or any two neutrons. Nevertheless, this statement furnishes a preliminary understanding of the actual conditions. This property of saturation of nuclear forces explains why—as already discussed—the binding energy of the individual building blocks of nuclei is independent of the size of the nucleus. If a particle becomes lodged in a nucleus, owing to the extremely short range of the nuclear forces it enters into interaction with its immediate neighbours only—quite unlike the case of electric forces—and secondly, owing to the saturation of the forces, this particle is capable of linking itself to two of these immediate neighbours only. This is a further justification of the analogy between nuclear matter and a liquid. For the same